TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	75.5 ± 2.33
Correlation Stability	89.0 ± 1.79
Distribution Consistency	85.0 ± 1.90
Entropy Preservation	87.3 ± 1.85
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	84.2 ± 0.60
Privacy Risk	50.3 ± 1.27
Data Density	100.0 ± 0.01
Coverage Score	22.9 ± 5.19
Feature Accuracy	78.5 ± 4.13
Novelty Score	0.4 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	39.843	-4.903	11.0%	Warn
TB	3.299	1.397	-1.902	57.7%	Drift
DB	1.486	0.525	-0.961	64.7%	Drift
Alkphos	290.576	206.415	-84.162	29.0%	Drift
Sgpt	80.714	36.230	-44.484	55.1%	Drift
Sgot	109.911	43.311	-66.600	60.6%	Drift
TP	6.483	6.764	+0.281	4.3%	OK
ALB	3.142	3.643	+0.501	15.9%	Warn
A/G Ratio	0.947	1.170	+0.223	23.6%	Warn
Selector	1.286	1.471	+0.184	14.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	75.5 ± 2.33
Correlation Stability (*2)	89.0 ± 1.79
Distribution Consistency (*3)	85.0 ± 1.90
Entropy Preservation (*4)	87.3 ± 1.85
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	84.2 ± 0.60
Privacy Risk (*7)	50.3 ± 1.27
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	22.9 ± 5.19

Feature Accuracy (*10)	78.5 ± 4.13
Novelty Score (*11)	0.4 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7136752136752136	0.4717948717948718	0.6888888888888889	-0.02478632478632479	
F1	0.8136759113636947	0.5261734404018433	0.7980365066885478	-0.01563940467514683	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.653846153846154	0.4982905982905982	0.6470085470085472	-0.00683760683760683	
F1	0.7583652592453195	0.5647964081516725	0.7455814675221417	-0.012783791723177607	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.4863247863247862 6	0.6435897435897435	-0.04273504273504273	
F1	0.7971299319768493	0.602068774327206	0.7660075830473104	-0.031122348929538968	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.4478632478632479	0.7051282051282051	-0.00085470085470085 17	
F1	0.8128002697439015	0.4975426803016908	0.801954455912713	-0.010845813831188444	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.452991452991453	0.5658119658119658	0.006837606837606857 6	
F1	0.5587432119380917	0.5047527732081885	0.568462110565521	0.009718898627429307	

Model: AB

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.4717948717948717	0.6743589743589744	0	
F1	0.7821343746683586	0.5084059545454958	0.7856321718765583	0.003497797208199744	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.4675213675213675	0.6760683760683761	-0.026495726495726513	
F1	0.7997903571549948	0.5178233547670361	0.7750108408925359	-0.024779516262458788	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.5615384615384615	0.7068376068376068	0	
F1	0.8279091624529537	0.6958864189549099	0.8279091624529537	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6803418803418804	0.4581196581196581	0.676923076923077	-0.0034188034188033954	
F1	0.7880117071652893	0.513918184665404	0.7805299976220242	-0.007481709543265075	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6837606837606838	0.4273504273504273	0.6760683760683761	-0.007692307692307709	
F1	0.7905911731742301	0.4719393240540625	0.7915898089374264	0.0009986357631958764	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	24	Male	3.3	1.6	174	11	33	7.6	3.9	1.0
2	42	Female	0.5	0.1	162	155	108	8.1	4.0	0.9
3	36	Male	0.8	0.2	182	31	34	6.4	3.8	1.4
4	34	Male	6.2	3.0	240	1680	850	7.2	4.0	1.2
5	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
6	51	Male	2.2	1.0	610	17	28	7.3	2.6	0.55
7	29	Female	1.2	0.5	190	35	40	6.8	3.5	1.1
8	34	Male	2.0	0.8	220	45	50	7.0	3.8	1.3
9	41	Female	0.9	0.3	160	25	30	6.5	3.2	1.0
10	55	Male	3.5	1.5	300	60	70	7.2	4.0	1.2
11	62	Female	1.8	0.7	210	50	55	6.9	3.6	1.1
12	38	Male	2.5	1.0	250	40	45	6.7	3.4	1.0
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	75	Female	0.8	0.2	205	27	24	4.4	2.0	0.8
2	68	Female	0.7	0.2	186	18	15	6.4	3.8	1.4
3	53	Male	19.8	10.4	238	39	221	8.1	2.5	0.4
4	12	Male	1.0	0.2	719	157	108	7.2	3.7	1.0
5	36	Male	5.3	2.3	145	32	92	5.1	2.6	1.0
6	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
7	29	Female	1.2	0.3	210	30	35	6.5	3.8	1.7
8	54	Male	0.9	0.2	180	40	50	7.0	3.5	0.9
9	45	Female	1.5	0.4	250	50	60	6.8	3.2	1.0
10	37	Male	2.0	0.5	300	45	55	7.2	3.0	1.2
11	60	Female	0.8	0.1	190	20	30	6.0	3.4	1.1
12	50	Male	1.4	0.3	220	35	40	6.7	3.6	1.0
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	26	Male	7.1	3.3	258	80	113	6.2	2.9	0.8
2	35	Male	0.8	0.2	279	20	25	7.2	3.2	0.8
3	53	Female	0.8	0.2	193	96	57	6.7	3.6	1.16
4	32	Male	0.7	0.2	276	102	190	6.0	2.9	0.93
5	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
6	64	Female	0.8	0.2	178	17	18	6.3	3.1	0.9
7	29	Female	1.2	0.5	210	35	30	6.8	3.5	1.2
8	45	Male	2.5	1.0	300	50	40	7.0	3.8	1.5
9	36	Female	0.9	0.3	180	22	25	6.5	3.2	1.0
10	52	Male	3.1	1.4	250	45	35	7.2	3.9	1.1
11	41	Female	1.5	0.6	220	30	28	6.9	3.6	1.2
12	34	Male	2.0	0.8	240	38	32	6.7	3.4	1.1
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.2	182	23	28	6.8	2.9	0.7
2	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51
3	55	Male	4.4	2.9	230	14	25	7.1	2.1	0.4
4	27	Male	1.3	0.6	106	25	54	8.5	4.8	nan
5	12	Male	1.0	0.2	719	157	108	7.2	3.7	1.0
6	65	Male	0.9	0.2	170	33	66	7.0	3.0	0.75
7	29	Female	0.6	0.3	150	32	25	6.9	4.0	1.2
8	34	Male	1.2	0.5	210	45	30	7.5	3.8	0.9
9	47	Female	0.9	0.4	180	28	22	6.5	3.5	1.0
10	52	Male	2.3	1.0	300	50	40	8.0	4.5	1.1
11	38	Female	1.0	0.2	160	35	20	7.0	3.9	0.8
12	45	Male	1.5	0.7	220	42	35	6.8	4.2	1.0
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
2	37	Male	1.8	0.8	145	62	58	5.7	2.9	1.0
3	61	Male	1.5	0.6	196	61	85	6.7	3.8	1.3
4	29	Male	1.2	0.4	160	20	22	6.2	3.0	0.9
5	42	Male	2.7	1.3	219	60	180	7.0	3.2	0.8
6	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
7	22	Female	0.5	0.1	150	30	25	6.5	4.0	1.6
8	45	Male	1.2	0.4	200	40	35	7.0	3.5	1.2
9	30	Female	0.8	0.2	180	25	20	6.8	4.2	1.5
10	50	Male	2.5	1.0	250	50	45	7.5	3.0	1.0
11	34	Female	0.9	0.3	220	35	30	6.7	4.1	1.4
12	29	Male	1.1	0.4	190	28	22	6.6	3.8	1.3
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	21	Male	0.7	0.2	135	27	26	6.4	3.3	1.0
2	33	Male	0.9	0.8	680	37	40	5.9	2.6	0.8
3	42	Male	0.8	0.2	158	27	23	6.7	3.1	0.8
4	60	Male	6.3	3.2	314	118	114	6.6	3.7	1.27
5	49	Male	3.9	2.1	189	65	181	6.9	3.0	0.7
6	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
7	29	Male	1.2	0.4	210	45	30	6.9	3.5	1.2
8	34	Female	0.9	0.3	180	40	25	7.1	3.8	1.5
9	42	Male	2.5	1.0	250	60	35	6.5	3.0	0.8
10	55	Female	1.8	0.7	220	50	40	7.4	3.6	1.0
11	38	Male	3.0	1.5	300	70	50	6.8	3.2	0.9
12	27	Female	0.6	0.2	160	30	20	7.0	4.0	1.3
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
2	36	Male	0.9	0.1	486	25	34	5.9	2.8	0.9
3	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
4	32	Male	15.0	8.2	289	58	80	5.3	2.2	0.7
5	60	Male	5.8	3.0	257	107	104	6.6	3.5	1.12
6	46	Male	0.7	0.2	224	40	23	7.1	3.0	0.7
7	29	Female	1.2	0.3	210	40	35	6.8	3.5	1.2
8	54	Male	0.9	0.2	180	25	30	7.0	3.2	0.9
9	38	Female	2.5	1.0	250	60	70	6.5	3.0	1.1
10	45	Male	1.5	0.5	220	55	50	7.2	3.8	0.8
11	60	Female	3.0	1.5	300	70	80	6.9	3.4	1.0
12	50	Male	0.8	0.1	190	20	25	6.0	2.8	0.7
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18	Male	0.7	0.1	312	308	405	6.9	3.7	1.1
2	34	Male	4.1	2.0	289	875	731	5.0	2.7	1.1
3	42	Male	1.0	0.3	154	38	21	6.8	3.9	1.3
4	46	Female	0.8	0.2	185	24	15	7.9	3.7	0.8
5	55	Male	3.3	1.5	214	54	152	5.1	1.8	0.5
6	31	Male	0.6	0.1	175	48	34	6.0	3.7	1.6
7	22	Female	1.2	0.3	210	45	30	6.5	3.8	1.2
8	29	Male	0.9	0.2	180	50	40	7.0	4.0	1.5
9	35	Female	2.5	1.0	300	60	50	7.5	3.5	0.9
10	41	Male	1.8	0.4	220	55	45	6.8	3.9	1.1
11	50	Female	3.0	1.5	400	70	60	8.0	4.5	1.3
12	27	Male	0.6	0.1	150	30	20	5.5	2.8	0.8
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	26	Male	42.8	19.7	390	75	138	7.5	2.6	0.5
2	28	Male	0.6	0.2	159	15	16	7.0	3.5	1.0
3	29	Female	0.7	0.1	162	52	41	5.2	2.5	0.9
4	35	Female	0.6	0.2	180	12	15	5.2	2.7	nan
5	53	Male	19.8	10.4	238	39	221	8.1	2.5	0.4
6	45	Male	0.6	0.1	196	29	30	5.8	2.9	1.0
7	34	Female	1.2	0.4	210	45	30	6.5	3.2	1.1
8	50	Male	2.5	1.0	300	60	50	7.0	4.0	0.8
9	29	Female	0.8	0.2	150	35	25	6.0	3.0	1.0
10	45	Male	1.5	0.6	220	55	40	7.5	3.5	0.9
11	38	Female	0.9	0.3	180	50	35	6.8	3.4	1.2
12	60	Male	2.0	0.8	400	70	60	8.0	4.5	0.7
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	2.2	0.8	209	25	20	8.0	4.0	1.0
2	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0
3	44	Male	0.9	0.2	182	29	82	7.1	3.7	1.0
4	45	Female	0.7	0.2	153	41	42	4.5	2.2	0.9
5	33	Male	1.2	0.3	498	28	25	7.0	3.0	0.7
6	60	Male	11.0	4.9	750	140	350	5.5	2.1	0.6
7	29	Female	1.2	0.4	210	35	22	6.5	3.2	1.2
8	54	Male	0.9	0.3	180	40	30	7.0	4.0	1.5
9	37	Female	1.5	0.5	250	50	45	6.8	3.5	1.0
10	45	Male	2.0	0.8	300	60	50	7.5	4.2	1.1
11	62	Female	3.1	1.0	400	70	60	5.9	2.8	0.9
12	48	Male	0.8	0.2	220	30	25	6.2	3.0	1.1
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3